

What Remains Useful After Strong Frontier Prompting? Complementary Local Models for Ultra-Low-Resource Tangut Title Translation

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Abstract

Tangut short-text translation combines two difficulties: the data are extremely scarce, and the targets are often compact historical titles rather than ordinary modern sentences. In our repository snapshot, only 491 real Tangut-Chinese pairs are available, which makes the central question sharper than in ordinary low-resource MT. Once a strong frontier Chinese LLM is already equipped with dictionary glosses and a title-oriented prompt, what remains worth learning locally? We answer this with a layered case study: first establish a deliberately strong frontier prompt-only upper bound, then compare synthetic local supervised fine-tuning (SFT) recipes and filtered direct preference optimization (DPO), and finally test whether frontier and local candidates can be combined through an open selector or a closed catalog-aware adjudicator. The pattern is consistent. Frontier prompting is the strongest single-model strategy. Within the open/local regime, structured multitask supervision is the right base, and DPO helps only after aggressive preference-gap filtering. Local training is still useful not because it replaces frontier prompting, but because it contributes conservative alternative titles that improve downstream selection. The broader lesson is to frame ultra-low-resource historical translation as workflow design under reference-aware evaluation, since lexical coverage and perplexity can mis-rank the gold reference itself.

1 Introduction

Historical translation at the ultra-low-resource end is difficult for a different reason than ordinary low-resource MT. The challenge is not only that parallel data are scarce, but also that the desired outputs are often compact titles, glosses, or catalog strings rather than fluent modern sentences. A system can therefore sound plausible while still failing at the real scholarly task, which is to recover the right historical title in the right form. Tangut is a sharp case. In the repository studied here, only 491 real Tangut-Chinese pairs are available, and many targets are short title-like strings rather than ordinary modern sentences [1, 2, 3].

These coupled constraints create two concrete risks. On the modeling side, local training can easily drift toward synthetic artifacts or explanatory paraphrase instead of compact title reconstruction. On the evaluation side, generic MT metrics can reward plausible semantic overlap while still missing the historical title itself.

This changes the paper’s central question. Once a strong frontier Chinese LLM is given dictionary glosses and a title-oriented prompt, the interesting issue is no longer whether a local model can simply beat it head-to-head. The more useful question is what remains worth learning locally, and whether local hypotheses can improve a scholar-facing workflow even when frontier prompting is already strong.

We therefore organize the paper around three research questions:

1. **RQ1:** How far can strong frontier prompting go as a single-model upper bound?
2. **RQ2:** If local training still matters, what kind of supervision signal helps most?
3. **RQ3:** Do frontier and local systems fail differently enough to yield a reproducible workflow gain?

We answer these questions by comparing local dictionary-grounded prompting, a deliberately strong frontier prompt-only baseline, synthetic pseudo-parallel SFT, preference optimization on top of SFT, and inference-time selection over frontier and local candidates [5, 4, 11]. The experimental design is intentionally layered: first establish the strongest single-model upper bound, then identify the best open/local recipe, and only then ask whether complementarity can be turned into a workflow gain. Because the test set contains only 50 items and the numerically strongest adjudicator is closed, we state that final claim conservatively: the deterministic open selector is the main reproducible result, while the closed adjudicator is used as an upper bound on recoverable complementarity rather than as the paper’s central deployable artifact.

The story that emerges is simple. Strong frontier prompting already sets a high single-model bar, so the paper cannot rely on a weak-baseline narrative. Local training still matters, but only when it is disciplined enough to preserve title structure: structured multitask supervision is the right base, and DPO helps only after both the base model and the preference pairs are cleaned aggressively. The main positive result is therefore not local replacement of the frontier system, but recovery of complementary conservative hypotheses through open selection, with closed adjudication used only to estimate remaining headroom. At the same time, lexical coverage and perplexity mis-rank the gold reference itself, so generic reference-free signals are unsafe as primary evidence.

The paper makes three contributions:

1. a research-question-driven case study that separates three claims often blurred together in low-resource work: the strongest single-model upper bound, the best open/local trainable recipe, and the extent to which complementary local hypotheses can be recovered in a workflow under only 491 real pairs;
2. a positive local-method result showing that structured multitask supervision is the right base, that DPO helps only after aggressive preference-gap filtering, and that local models matter because they contribute complementary conservative hypotheses rather than because they dominate the frontier model head-to-head;
3. a reference-aware evaluation and label-quality analysis showing why short historical titles must not be judged primarily by generic reference-free metrics, why low-gap preference data destabilize alignment, and how much complementarity an open selector can recover reproducibly.

The rest of the paper follows the research questions in that order and keeps two final claims separate on purpose: how to build the strongest open Tangut model, and how much scholar-facing benefit can be recovered from frontier/local complementarity without overclaiming beyond the evidence.

2 Related Work

Ancient and historical language processing. Ancient-language NLP differs from mainstream NLP not only because data are scarce, but also because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on historical translation and document

modernization report similar sparsity, though they often target broader sentence translation or orthographic updating rather than cross-script title reconstruction [2, 3]. Our Tangut setting is narrower still: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Low-resource MT, synthetic supervision, and lexical grounding. Low-resource MT often relies on synthetic parallel data, multilingual transfer, or explicit lexical constraints [4, 6, 8, 5, 7]. Tangut overlaps with this literature but is harsher in two ways: the source script is largely outside standard MT pretraining, and the outputs are compact catalog titles rather than ordinary sentences. We therefore focus on synthetic supervision and lexicon-grounded structure rather than on broad multilingual transfer.

Dictionary-structured prompting and error-aware prompting. Recent LLM translation work suggests that lexical resources are most useful when presented as structured intermediate evidence rather than as flat prompt context. Chain-of-Dictionary prompting and Learning-From-Mistakes prompting both follow this pattern in low-resource settings [9, 10]. Our stronger frontier baseline is adjacent to this line, but the present paper asks a different question: whether alignment structure should be absorbed into trainable local targets, and whether those local hypotheses remain useful after a strong frontier prompt-only baseline is already in place.

Preference optimization. Direct Preference Optimization (DPO) is now a standard offline alignment objective, and recent work has extended it to machine translation and noisy-preference settings [11, 14, 15, 12, 13]. Our results place this literature in a harsher regime: supervision is tiny, outputs are title-like, and the chosen/rejected signal is built from partially misaligned automatic metrics. In this setting, DPO helps only after base-model choice and pair filtering are controlled aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under orthographic variation [16]. Learned MT metrics and QE systems such as COMET and OpenKiwi go beyond surface overlap, but recent work also shows that they can have domain-specific failure modes and induce metric interference [17, 20, 18, 19]. Our findings sharpen that caution in a historical short-text regime: lexical coverage and perplexity can penalize the gold reference itself, so reference-aware evidence must remain primary.

3 Task Setting

The paper studies Tangut short-text translation under two coupled constraints. The first is supervision scarcity: the real data in this repository contains only 491 pairs, split into 400 training examples, 41 development examples, and 50 test examples by a single random shuffle with seed 42. The second is output mismatch: unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many resemble book titles, sutra titles, or compact scholastic expressions, so success depends on precise lexical choices and compact formatting rather than on generic sentence fluency. These same constraints create the two failure modes addressed in the rest of the paper: local models can overfit noisy supervision and over-generate, while generic MT metrics can reward plausible paraphrase rather than exact title recovery.

This means the task is narrow but meaningful. We ask how to recover short historical titles under only 400 real training examples, not how to solve open-domain Tangut translation. To train local models at all, we therefore use synthetic data as structured proxies. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples upsampled by a factor of 10 to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone. These synthetic inputs are not claims about faithful Tangut reconstruction; they are judged only by downstream utility on held-out real Tangut titles.

We also audit data hygiene conservatively. There is zero exact source, target, or full-pair overlap across the train/dev/test splits, zero exact overlap between the 50 test references and the final targets of the 50k synthetic multitask corpus, and zero exact overlap between the frontier few-shot exemplars and the test references. This does not prove the absence of all leakage, but it rules out the simplest exact-copy explanation for the reported gains.

Evaluation is the second bottleneck, so we separate primary evidence from supporting diagnostics. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [16]. Perplexity is computed as $\exp(\ell)$, where ℓ is the mean token-level causal-LM loss under Qwen2.5-0.5B; in this title-heavy setting it becomes extremely large even for the gold reference, so we treat it as diagnostic rather than dispositive. “Contamination” in the current paper means mixed-script or artifact contamination in the *output* string, such as Latin letters, literal [UNK] tokens, or prompt residue, not train-test data leakage. Learned MT metrics and QE systems can capture richer signals than surface overlap [17, 20], but they can also have domain-specific failure modes [18, 19]. We therefore do not use off-the-shelf COMET as a headline arbiter: standard checkpoints are designed for modern sentence-level MT rather than Tangut short-title reconstruction. Accordingly, chrF++, exact match, contamination, suffix preservation, and the reference-aware judge are treated as primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge remain diagnostic.

4 Methods

We design the comparison around the research questions rather than around every implementation variant. The full ablation matrix remains in the repository; the main text keeps only the families needed to answer whether strong prompting is already enough, what local training signal remains useful, and whether frontier/local complementarity can be turned into a workflow.

4.1 Prompt-only upper bounds

We start with the strongest version of the prompt-only story. The local prompt baselines contain no weight updates: **baseline1** prompts the local model directly on the Tangut string, **baseline2** adds dictionary glosses, and **baseline2_1_cot** further decomposes generation into alignment, literal drafting, and rewriting. This family is closest to lexicon-augmented MT, except that the lexicon is injected through prompting rather than decoder constraints [5, 7].

We then add a deliberately strong frontier prompt-only baseline based on DeepSeek-V3.2 with dictionary grounding and four fixed development-set exemplars spanning canonical, bodhisattva-practice, dharani-style, and transliteration-heavy titles. The prompt emphasizes compact title form, forbids explanatory paraphrase, and emits only the final title string. Decoding uses temperature 0.0 with a 512-token cap. The exact prompt script and exemplar IDs are stored in the repository. The purpose is to make the objection as strong as possible: if frontier prompting already solves the task, the local-training story should shrink accordingly.

4.2 Local training recipes

If prompting is not the whole story, the next question is what kind of local supervision helps most. We build 50k pseudo-Tangut/Chinese pairs from monolingual historical Chinese by mapping Chinese characters to Tangut characters through dictionary explanations, fixing the synthesis seed to 42, and keeping source lengths between 4 and 64 characters. The four SFT branches differ only in how dictionary misses are handled and whether the target includes alignment structure: mixed-script keeps 30–70% Chinese characters, UNK replaces misses with [UNK], semantic projection replaces misses through Qwen2.5-0.5B embeddings plus FAISS lookup, and multitask keeps the UNK input but changes the target to `<dict_match>{...}</dict_match> <literal>{...}</literal> {final title}`. This gives the model both an alignment channel and a final title target in one sequence. These examples are not philologically faithful Tangut reconstructions; they are structured training proxies. Each synthetic set is combined with the 400 real training examples upsampled by a factor of 10 and shuffled with seed 42.

In the narrative that follows, two local variants matter most. UNK-SFT isolates the effect of input purity, while multitask SFT tests whether explicit alignment structure adds value beyond purity alone. The mixed and semantic variants remain part of the repository-level ablation matrix, but they do not change the main story.

4.3 Preference alignment on local bases

The DPO family asks a narrower question: once a strong local base exists, can reward-based alignment improve it without damaging title form? We apply DPO to an SFT base using reward-defined chosen/rejected pairs [11]. We sample five outputs per training prompt at temperature 0.8 and top- p 0.95, then score them with

$$r(y) = \text{lex}(y) - 0.01 \log \text{ppl}(y),$$

where $\text{lex}(y)$ is the repository’s lexical-coverage score. The key point is not the exact tokenizer details, but that the reward favors dictionary consistency with only a weak fluency tie-breaker. We then keep the top-scoring output as chosen and the bottom-scoring output as rejected whenever the reward gap is at least 0.05.

The legacy pipeline first selects the best SFT base among the mixed, UNK, and semantic variants using chrF++, then constructs preference pairs from that base. In the current snapshot that selector chooses the UNK model, yielding the legacy checkpoint `final_v2`, which serves as a negative control. The follow-up DPO runs instead start from the stronger multitask SFT model. After auditing the preference data, we keep only pairs whose reward gap exceeds either 0.2 (1,996 pairs) or 0.4 (498 pairs), and we train both standard sigmoid DPO and a robust+weighted variant [12, 13]. This turns the DPO family into a controlled test of base-model choice, pair filtering, and loss robustness.

4.4 Hybrid adjudication

The final method family operationalizes the paper’s main positive claim. The oracle analysis suggested that frontier and local candidates are complementary, but an oracle is not a deployable system. We therefore build a real inference-time adjudicator over the frontier DeepSeek baseline, UNK-SFT, and MT-DPO g0.4 sig. For each item it sees the Tangut source, dictionary glosses, the three candidate titles, their lengths, and simple prefix/suffix consensus hints.

The adjudicator uses Azure-hosted GPT-5.4 at temperature 0.0. Its prompt treats the task as historical catalog-title reconstruction rather than free paraphrase: dictionary glosses are clues

rather than canonical modern titles, and short inputs are penalized for explanatory expansion. The model may either copy one candidate or perform a minimal repair using only candidate fragments or directly attested gloss segments. The repository stores the exact templates, frontier prompting script, fixed exemplar IDs, and per-example adjudication JSON, so the closed-model part of the workflow is at least auditable.

For reproducibility, we also add a deterministic open selector over the same 3-way pool. It never synthesizes text or calls an external model. It uses title-suffix priors, candidate-length agreement, contamination penalties, conservative guards against truncation or expansion, and a small frontier prior. A guarded switch replaces the frontier output only when a local candidate has materially stronger conservative evidence.

4.5 Implementation details

All learned models start from Qwen2.5-7B-Instruct with LoRA adapters so that differences are attributable to supervision strategy rather than backbone changes. SFT uses three epochs, learning rate 2×10^{-4} , batch size 4, gradient accumulation 4, rank-64 LoRA, and maximum sequence length 512. DPO uses two epochs, learning rate 5×10^{-5} , batch size 1, gradient accumulation 8, $\beta = 0.1$, rank-32 LoRA, cosine decay, and the same sequence length; the robust variants add label smoothing 0.1 and WPO-style weighting. The full pipeline fixes seed 42 for synthesis, data combination, and DPO training.

The reference-aware judge is a supplementary evaluator, not a training signal. It uses Azure-hosted GPT-5.4 to score reference agreement, source faithfulness, title-style fitness, and overall quality on a 1–5 scale after seeing the Tangut source, gold reference, and candidate simultaneously. For cross-judge robustness, the repository also includes a local Qwen2.5-7B-Instruct judge with the same rubric. We use the Azure judge as the primary table because it is the stricter and less saturated scorer.

5 Main Results

To keep the paper centered on the research questions, Table 1 reports only representative systems. The full ablation matrix, including zero-shot prompting, mixed and semantic SFT, gap-0.2 DPO, robust DPO, and broader workflow variants, remains in the repository and does not change the conclusions. The empirical story is short: a strong frontier baseline already answers much of the single-model question, local learning helps only when it stays close to title structure, and the main reproducible gain comes from selecting among complementary candidates rather than asking one local model to replace the frontier system.

5.1 RQ1: Is strong frontier prompting already enough as a single-model upper bound?

Yes for the single-model question. The DeepSeek frontier baseline is far stronger than the local prompt baselines, reaching 28.54 chrF++, 12/50 exact matches, and 2.80 overall under the reference-aware judge. The important consequence is interpretive as much as empirical: prompt-only is not a straw baseline here. A strong frontier system is already hard to beat, so every later claim has to explain what local training still contributes after that bar is in place.

Method	chrF++ $\uparrow$	Exact Match $\uparrow$	Ref.-Aware Overall $\uparrow$
<i>Prompt-only</i>			
Dict prompt	6.13	0/50	1.50
<i>Frontier DeepSeek</i>	28.54	12/50	2.80
<i>Local learned</i>			
UNK-SFT	22.11	3/50	1.82
MT-SFT	25.99	5/50	1.98
Legacy DPO	19.39	2/50	1.68
<i>MT-DPO g0.4 sig</i>	30.01	5/50	2.22
<i>Hybrid workflow</i>			
<i>Open selector (3-way, open)</i>	32.62	12/50	2.90
Catalog adjudicator (3-way, closed)	34.87	14/50	3.02
Human ref.	100.00	50/50	5.00

Table 1: Representative systems aligned with the paper’s first three research questions. Frontier DeepSeek is the strongest single model, MT-DPO g0.4 sig is the strongest local learned model, and the open 3-way selector is the main reproducible workflow result. The closed 3-way catalog adjudicator is reported as an upper bound on recoverable complementarity. The full ablation matrix remains in the repository and does not change these conclusions.

5.2 RQ2: If local training still matters, what signal helps most?

The answer is not that more local training is always better. What helps is supervision that explicitly teaches the model how a short historical title is assembled. UNK-SFT already improves over prompt-only local decoding by enforcing pure Tangut inputs, but multitask SFT improves again, making structured supervision the right base. DPO is not uniformly helpful: the legacy DPO checkpoint trained on the old UNK-selected base drops to 19.39 chrF++ and 1.68 overall, so reward-based alignment is clearly not a blanket win. Once DPO is moved onto the multitask base and filtered aggressively, however, it becomes useful: MT-DPO g0.4 sig is the strongest local model at 30.01 chrF++ and 2.22 overall. The omitted gap-0.2 run is much worse at 20.93 chrF++, showing that label filtering, not DPO alone, drives the gain.

5.3 RQ3: Can complementary failures be turned into a stronger workflow?

Yes, and this is where the paper’s main positive result appears. The frontier model remains the best single predictor, but the best local models are still useful because they fail differently and often more conservatively on short titles. A deterministic open selector over frontier, UNK-SFT, and MT-DPO g0.4 sig already reaches 32.62 chrF++ and 2.90 overall, so part of that complementarity is recoverable without closed adjudication. A 3-way catalog-aware adjudicator goes further to 34.87 chrF++, 3.02 overall, and 14/50 exact matches, but we treat that result as a closed upper bound on recoverable complementarity rather than as the paper’s main operational claim.

Two supplemental controls sharpen this claim. First, pool ablation shows that each local candidate family contributes real but limited complementarity on its own: frontier+UNK, frontier+MT-SFT, and frontier+DPO each repair one additional exact match and raise chrF++ from 28.54 to 30.05, while the full 3-way open selector reaches 32.62 chrF++, indicating that the larger gain is distributed across local candidates rather than explained by a single pairwise fix. Second, a targeted sensitivity sweep over frontier prior, title-suffix bonus, and guarded switch margin keeps the 3-way selector between 32.33 and 33.48 chrF++ with 11–12 exact matches and zero contamination,

Workflow	Pool	Exact Match	chrF++	Overall
<i>Frontier DeepSeek</i>	single	12/50	28.54	2.80
<i>3-way open selector</i>	F+UNK+DPO	12/50	32.62	2.90
3-way catalog adjudicator (closed)	F+UNK+DPO	14/50	34.87	3.02
Best-of-3 oracle	F+UNK+DPO	14/50	41.85	3.12

Table 2: Consolidated workflow comparison. “F” = frontier DeepSeek, “UNK” = UNK-SFT, and “DPO” = MT-DPO g0.4 sig. The deterministic selector is fully open and never edits text. The closed catalog adjudicator is reported as a stronger but less reproducible upper bound, while the oracle shows remaining headroom inside the same 3-way candidate pool.

so the open gain is not a brittle one-setting artifact.

The workflow result should therefore be read in two layers. Operationally, the open selector is the main reproducible gain. Analytically, the closed adjudicator shows that the candidate pool contains additional complementarity that a stronger reranker can exploit. The core claim is intentionally modest: frontier prompting is the single-model upper bound, MT-DPO g0.4 sig is the strongest local model, and local training matters because it supplies complementary conservative hypotheses that even an open selector can exploit.

6 Analysis

6.1 Why the workflow gain is conservative rather than verbose

Table 3 summarizes the output-side diagnostics. Here “contamination” flags outputs containing ASCII spillover, angle brackets, [UNK], or known prompt-artifact tokens, and “suffix preservation” is computed on references containing at least one canonical title-suffix character; a prediction is counted as correct when it preserves all such reference-side suffix characters. The table caption gives the exact suffix list.

The frontier baseline is already a clean single model: 12/50 exact matches, 0 contamination, 24/27 suffix preservation, and a near-reference length ratio of 0.95. Within the local family, multi-task SFT is the cleanest pure SFT system, and strict gap-0.4 DPO regularizes the outputs further. The 3-way open selector stays clean as well, while the 3-way catalog hybrid is cleaner still with 14/50 exact matches, 26/27 suffix preservation, 0 contamination, and the same 0.95 length ratio as the frontier baseline. The gain therefore does not come from verbose rewriting; it comes from better conservative title reconstruction.

6.2 Why three candidates are enough

The oracle analysis mattered because it showed that frontier and local candidates were not near-duplicates. In the paper’s interpretation, the open selector is the main reproducible workflow result, while the closed catalog adjudicator acts as a probe of how much additional complementarity is present in the candidate pool.

The 3-way deterministic selector over frontier/UNK-SFT/MT-DPO already changes 12/50 cases without any closed model or text synthesis: it selects the frontier 38 times, UNK-SFT 8 times, and MT-DPO 4 times. The 3-way catalog adjudicator goes further. It selects the frontier 36 times, MT-DPO 6 times, UNK-SFT 5 times, keeps one tie, and performs only 2 minimal repairs. Thus 48/50 outputs come from direct candidate selection rather than free rewriting, and local models contribute directly on 11 cases plus two repairs. The exact-match union of the three candidate

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
<i>Prompt-only</i>				
Dict prompt	0/50	0.00	3/27	1.83
<i>Frontier DeepSeek</i>	12/50	0.00	24/27	0.95
<i>Local learned</i>				
UNK-SFT	3/50	0.06	17/27	1.05
MT-SFT	5/50	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
<i>MT-DPO g0.4 sig</i>	5/50	0.04	22/27	1.12
<i>Hybrid workflow</i>				
<i>Open selector (3-way, open)</i>	12/50	0.00	23/27	0.98
Catalog adjudicator (3-way, closed)	14/50	0.00	26/27	0.95
Human ref.	50/50	0.00	27/27	1.00

Table 3: Representative reference-aligned diagnostics derived from stored predictions. “Contam.” denotes mixed-script or prompt-artifact contamination. “Suffix Preserve” is computed on references containing at least one title-suffix character in {經論記疏頌儀義傳錄贊序字品} and marks a prediction correct when it preserves all such reference-side suffix characters. The table is intended to show that the workflow gain comes from cleaner conservative title reconstruction rather than from verbose rewriting.

sources is also 14/50, so the closed adjudicator closes the exact oracle gap completely even though the oracle still retains headroom on chrF++ and overall score.

The new pool-ablation control clarifies why this is a real complementarity story rather than a single lucky override. When we pair the frontier model with only one local candidate at a time, frontier+UNK, frontier+MT-SFT, and frontier+DPO all land at the same intermediate operating point: 30.05 chrF++, 13/50 exact matches, and zero contamination. The full 3-way selector then pushes chrF++ further to 32.62, even though exact match returns to 12/50. This pattern matters because it shows that the open selector is not just finding one hidden exact-match repair; it is combining multiple conservative local alternatives to improve aggregate title reconstruction more broadly.

Additional pool-expansion variants did not improve the conservative workflow and are therefore left out of the main narrative. The qualitative cases make the core point. The selector restores 五部經 instead of the frontier normalization 五類經, fixes the frontier truncation 番 to 番言金剛王乘根, and replaces the frontier paraphrase 德宜點義句顯宣論 with the more title-like 正理滴之句義顯了. In other words, the gain is not that the system writes more, but that it sometimes knows when to stay closer to historical title structure.

6.3 Why reference-aware evaluation is necessary

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate simultaneously.

Table 4 changes the ranking in the direction the task actually demands. Among single models, the frontier baseline remains the clear winner at 2.80 overall. Within the local learned family, MT-DPO g0.4 sig is strongest overall at 2.22. Among workflows, the open selector already reaches 2.90 overall and the closed 3-way catalog adjudicator 3.02. The gains are concentrated where catalog conservatism matters most: on non-suffix titles the frontier baseline scores 2.39 overall and the

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
<i>Prompt-only</i>				
Dict prompt	1.50	1.52	2.10	1.50
<i>Frontier DeepSeek</i>	2.78	2.82	4.48	2.80
<i>Local learned</i>				
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
<i>MT-DPO g0.4 sig</i>	2.12	2.26	4.60	2.22
<i>Hybrid workflow</i>				
<i>Open selector (3-way, open)</i>	2.82	2.88	4.50	2.90
Catalog adjudicator (3-way, closed)	2.94	3.00	4.72	3.02
Human ref.	5.00	5.00	5.00	5.00

Table 4: Representative reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously, so this table is closer to the true task than the legacy reference-free judge.

Comparison ($B - A$)	Δ chrF++	Δ Overall	Δ Exact
Closed 3-way – Frontier	+6.33 [2.10, 11.67]	+0.22 [0.00, 0.48]	+0.04 [0.00, 0.10]
Open 3-way – Frontier	+4.08 [−2.28, 10.64]	+0.10 [−0.20, 0.40]	0.00 [−0.08, 0.08]
MT-DPO g0.4 sig – MT-SFT	+4.02 [0.84, 7.80]	+0.24 [0.00, 0.46]	0.00 [0.00, 0.00]
MT-DPO g0.4 sig – Frontier	+1.47 [−8.09, 10.29]	−0.58 [−1.04, −0.12]	−0.14 [−0.26, −0.04]
Legacy DPO – MT-SFT	−6.60 [−11.84, −1.73]	−0.30 [−0.64, 0.04]	−0.06 [−0.14, 0.00]

Table 5: Paired bootstrap comparisons with 5,000 resamples. “Overall” denotes mean reference-aware judge score, and “Exact” denotes exact-match rate. The selected rows correspond to the paper’s core claims: the open and closed workflow gains over frontier, the local DPO gain over MT-SFT, and the failure of legacy DPO.

3-way adjudicator 2.74, and on references of length ≤ 6 the frontier scores 2.94 and the adjudicator 3.24. A local Qwen2.5-7B judge preserves the same frontier/open/hybrid/human ordering (4.20, 4.20, 4.24, 4.90), reducing concern that the ranking is purely a GPT-family artifact.

The `human_reference` run is an important anchor. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, but only moderate lexical coverage and catastrophic perplexity. Lexical coverage therefore measures overlap with dictionary-derived semantics rather than agreement with the gold title, while perplexity penalizes compact title style and historical transliterations. These metrics remain useful diagnostics, but they should not drive model selection in this task.

6.4 Why filtered DPO changes the local story

Table 5 summarizes the robustness checks around the local and workflow claims.

The bootstrap results clarify which claims are stable. The local chrF++ gain from multitask SFT to strict gap-0.4 sigmoid DPO remains above zero under resampling, and the catalog hybrid’s chrF++ gain over the frontier baseline does as well. The judge-overall and exact-match gains are directionally positive but boundary-level on this 50-example test set. The overlap audit addresses a separate concern: there is zero exact source, target, or full-pair overlap across train/dev/test,

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap (0.05, 0.10]	62.3%
Chosen closer on gap (0.20, 0.40]	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 6: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check, which helps explain why strict filtering stabilizes the later multitask-base DPO runs.

zero exact overlap between test references and the 50k synthetic multitask targets, and zero exact overlap between the frontier few-shot exemplars and the test references. This does not prove complete immunity from leakage, but it rules out the simplest exact-copy explanation.

The DPO audit then explains why the local story changes once the labels are cleaned. Across all 3,867 preference pairs, the reward-chosen candidate is closer to a simple gold proxy only 71.3% of the time, but this rises to 81.5% for gaps ≥ 0.40 . The high-gap filter is therefore isolating cleaner labels rather than acting as a cosmetic hyperparameter. The reward-weight ablation sharpens the same conclusion: varying the coefficient on log ppl from 0 to 0.02 causes zero pair-order flips relative to the current reward and changes the gap-0.4 pool only from 498 to 504 pairs. In other words, the small perplexity term is not selecting different high-gap pairs; the real stabilizer is still aggressive gap filtering on top of the multitask base. Once DPO is moved onto that base and filtered at 0.4, it stops behaving like a blanket negative result and becomes the strongest local model. The hybrid workflow does not erase this point; it depends on the local candidate pool being strong enough to contribute conservative alternatives in the first place.

The selector sensitivity study provides a parallel robustness check on the workflow side. Across guarded 3-way configurations with frontier prior varied from 0 to 1.5, title-suffix bonus from 1.5 to 3.5, and switch margin from 2 to 4, the open selector stays within 32.33–33.48 chrF++, with 11–12 exact matches, 23–24 suffix preservations, and zero contamination. A more conservative margin-4 variant slightly increases chrF++ to 33.48 and restores suffix preservation to 24/27, but its reference-aware overall score drops from 2.90 to 2.86. We therefore keep the default selector as the main operating point: the open gain is robust across nearby settings, and the chosen default better balances overlap with judged title quality.

7 Discussion and Limitations

This paper should not be read as a collection of ablations in search of the highest score. It is a question-driven case study of what changes once a strong frontier model is already available under extreme scarcity. Under only 491 real pairs, the remaining scientific question is not whether a local model can beat that frontier system outright, but what kind of local signal is still worth learning. In this repository, the answer is structural conservatism: local models trained with explicit alignment scaffolding are more willing to stay close to short historical title structure, which makes them useful

complements even when they are not the best standalone systems.

This changes how the contribution should be framed. The value of local training is not that it beats a frontier API across the board. Its value is that it diversifies the candidate set with hypotheses that fail differently from frontier prompting. For scholar-facing historical translation, that is often the more realistic goal: produce a small pool of plausible, conservative titles and then choose among them with a title-aware reranker rather than a generic fluency prior. In that sense, the paper’s most reproducible positive claim is the open-selector result, now strengthened by explicit pool-ablation and parameter-sensitivity controls. The stronger closed adjudicator should be read as evidence that additional complementarity exists, not as the main deployable artifact.

The scope remains narrow. The current data support Tangut short-text translation, especially title-like strings, not open-domain sentence translation. The synthetic corpus is a structural proxy rather than naturally occurring pseudo-parallel data, and the strongest gains are concentrated on short and mid-length titles. The adjudication results are therefore best read as evidence about workflow design under extreme scarcity, not as a claim that closed-model reranking solves historical translation in general.

Two limitations are central. First, the dataset is tiny, so some judge and exact-match gains remain borderline even when the direction is stable. Second, the strongest adjudicator and reference-aware judge are closed models, and we do not yet have blinded expert validation. The next steps follow directly from those limits: build an open local reranker from the released adjudication traces, validate the judge on a small expert-annotated subset, and test whether the same “frontier upper bound + conservative local complement” pattern transfers to other historical short-title datasets.

8 Conclusion

Tangut short-text translation is difficult here not only because the dataset is tiny, but because the desired outputs are short historical titles that generic MT objectives and metrics do not model well. We therefore asked a narrower question: after strong dictionary-grounded frontier prompting is already in place, what remains useful about local training? The answer is that frontier prompting is the single-model upper bound, while structured local training remains valuable because it supplies conservative complementary titles that an open selector can exploit. The broader lesson is to frame ultra-low-resource historical translation as workflow design under reference-aware evaluation rather than as a head-to-head attempt to make one local model beat a frontier API.

The immediate next step is to turn the closed adjudication signal into stronger open reranking and to validate the workflow on a small expert-annotated subset.

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